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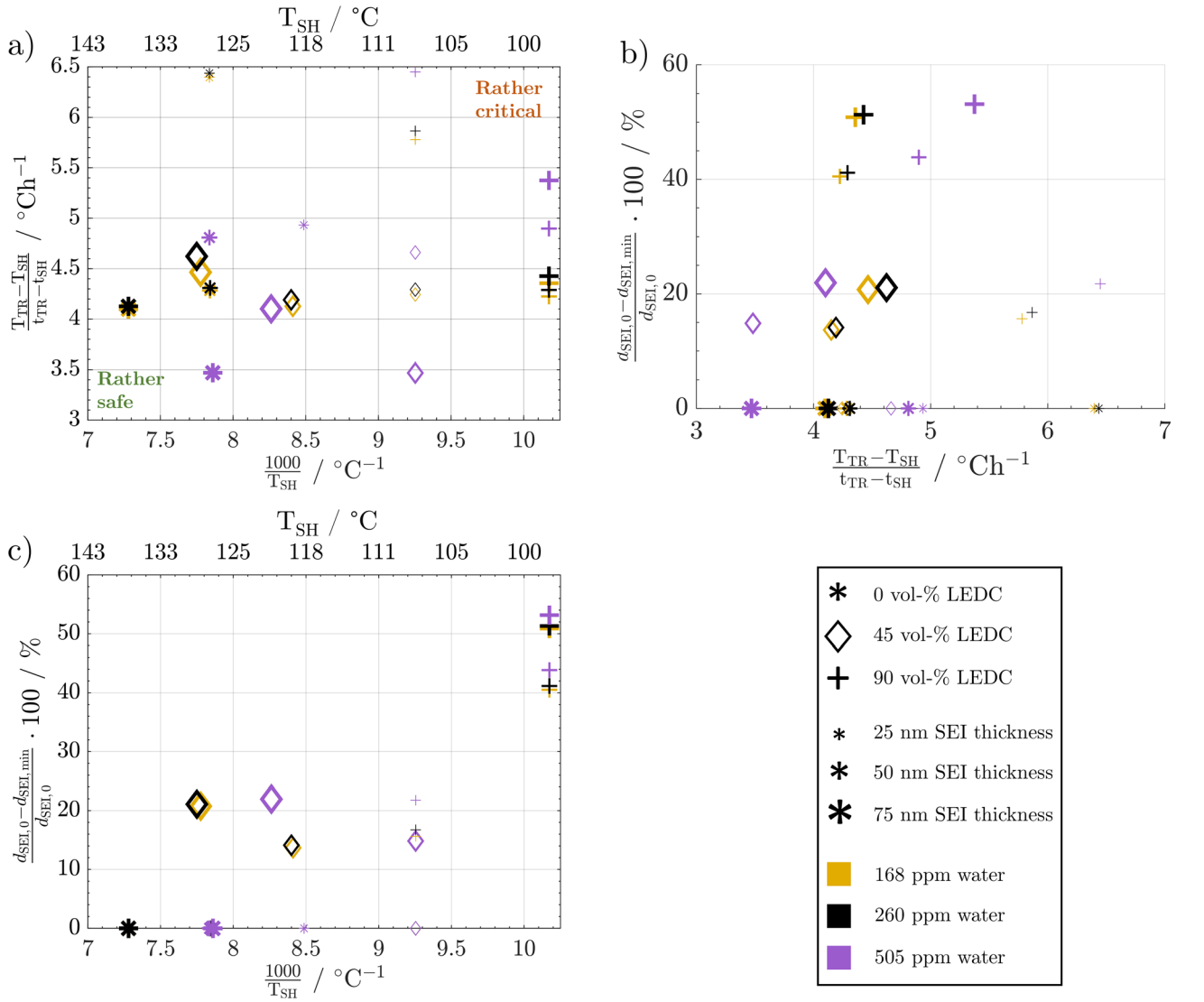

Fig. 8 Impact of changes in water content and SEI properties on: (a) temperature gradient over the inverse self-heating temperature scaled by 1000, (b) the percentage of SEI thickness reduction over the temperature gradient, and (c) the percentage of SEI thickness reduction over the inverse self-heating temperature scaled by 1000.

LEDC contents and a thin to medium thick SEI. The safe region on the other hand embodies almost exclusively all thick SEI variations with a large proportion of the inorganic cases. We can also observe that the thin SEI cases tend to have high temperature gradients probably due to the faster reformation at higher temperature. Higher LEDC content correlates with lower self-heating temperatures, as LEDC decomposition already occurs at low temperatures, whereas Li_2CO_3 decomposition is less strong at these temperatures. Variations of H_2O are scattered over the whole figure, which indicates that its influence is not as significant and straight-forward as the SEI properties even for high contaminations. This confirms the lower sensitivity to H_2O content than to SEI thickness and LEDC content.

The complex water effects warrant further analysis. For cells with thick inorganic SEIs (large stars, Fig. 8a) the wet case (violet) has a lower self-heating temperature and a lower gradient compared to the dry and medium dry case. This means that it enters the self-heating phase earlier but then heats up slower, *i.e.* one metric gets worse and the other gets better. In contrast, for the high LEDC contents (+) the higher water contamination value does not lead to lower self-heating temperature and the temperature gradient shows the opposite behaviour to the inorganic case: it increases. The root cause lies in the onset of LiOH decomposition. Only in cases of high water contamination is the LiOH amount high enough to significantly impact the temperature progression, which we showed in Fig. 7a. The onset of this exothermic decomposition is around >115 °C, which is higher than the self-heating temperature of all high LEDC cases. This means that the LEDC cases are already in sustained self-heating when the LiOH decomposes; thus, water content does not impact the self-heating temperature for high LEDC content cases, whereas for all cases that have not transitioned to sustained self-heating before LiOH decomposes, the heat from LiOH decomposition impacts the self-heating

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temperature: the higher the water content and thus LiOH concentration, the more LiOH decomposition heat, thus the lower the self-heating temperature. The observed lower temperature gradient for the wet, inorganic and reference LEDC concentration cases is directly correlated with the lower temperature itself. The lower the temperature the slower all reactions progress, and thus, the less heat is produced.

A low SEI thickness has been shown to accelerate the exothermic SEI reformation; thus, we analyse for the different cases how much the SEI is reduced and how this correlates to the temperature gradient (Fig. 8b) and the self-heating temperature (Fig. 8c). The high LEDC content cases (+) exhibit the highest decline in SEI thickness compared to the medium and inorganic cases. The low thicknesses for high LEDC contents in turn accelerate the reformation and, thus, contribute to higher temperature gradients. The high concentration of LEDC also leads to a low self-heating temperature. The pure inorganic cases, represented by stars, and most of the thin SEI cases, represented by small symbols, do not show any thickness reduction. Inorganic SEI cases do not show a decrease in SEI thickness because the decomposition of Li_2CO_3 is not substantial enough to compensate for the SEI formation rates even for the thick SEI cases. Therefore, no decrease in SEI thickness can be observed. For the thin SEI cases, the thickness does not decrease substantially because the thin SEI accelerates SEI reformation, which counteracts SEI decomposition reactions. Most of the thin SEI and the inorganic SEI cases show a correlation between the temperature gradient, the self-heating temperature and the SEI thickness: the thinner the SEI the higher the temperature gradient and the lower the self-heating temperature. For the high LEDC content and thin SEI cases, the high amount of LEDC decomposes before substantial reformation starts. Thus, they show a slight SEI thickness reduction. Due to their anyway thin SEI thickness they also show the highest temperature gradient among all organic cases. Comparing the thin SEI cases with the medium thick SEI cases it shows that the medium thick SEI cases exhibit lower temperature gradients and higher self-heating temperatures. This is explained by the accelerated reformation rates for the thin SEI cases, shereas, the comparison of the medium thick SEI and thick SEI cases reveals that the thick SEI cases have a higher temperature gradient. This is explained by the higher self-heating temperature of the thick SEI cases. Since they transition at higher temperatures into the self-heating phase, the reactions occur faster and more heat is produced.

From these extensive variations and their impact on thermal safety behaviour, it can be concluded that the SEI thickness and LEDC content are the dominant effects in terms of battery safety. A safer battery has an inorganic, thick SEI. Safety decreases with increasing LEDC content and reducing SEI thickness. Besides these two, H_2O impurities only play a role when severe contamination is present and generally contribute less than the SEI properties. The effect of water is also more challenging to address. High contamination affects the self-heating temperature and the temperature gradient differently depending on the composition of the SEI.

4 Conclusion

This study has elucidated the impact of the initial SEI state and water impurities on the thermal safety behaviour of Li-ion batteries with EC:EMC 1.2 M $LiPF_6$ and a graphite anode, including when and why self-heating occurs and the subsequent progression to rapid thermal runaway. Initial concentrations of SEI components, impurities and conductive salt decomposition products were rigorously derived from assessing manufacturing, production and ageing effects.

Dominant detrimental effects are high LEDC concentrations and a thin SEI, such as those found in rather fresh cells. Here, a high LEDC content could be connected to an earlier onset of self-heating. In contrast, a thinner SEI relates to faster SEI reformation and thus to a higher temperature gradient. The experimentally observed increase in self-heating temperature for aged cells[20–23] is thus attributed to an ageing-induced change from a foremost organic SEI to an inorganic SEI and a thicker SEI, which delays exothermic SEI reformation processes. The impact of H_2O impurities on battery safety is found to be marginal as long as a moderate drying procedure is applied to the electrodes during manufacturing. Thus, we could show that extreme electrode drying does not benefit battery safety. However, high H_2O contamination during production should be avoided as this will have a substantial negative impact. Here, the effect of high contamination was found to depend on the SEI composition. For an inorganic and mixed SEI, the contamination will reduce the self-heating temperature due to decomposition of LiOH. High LEDC content cases, on the other hand, exhibit a higher temperature gradient, because here the self-heating already starts before LiOH decomposition sets in.

The insights gained here contribute significantly to understanding and controlling Li-ion battery behaviour during thermal abuse. The trends for impact of water as an electrolyte impurity, the complexities of SEI properties, and their combined battery safety have been shown, and they cross influence each other. The presented degradation reactions and kinetics are suitable for integration into full cell models to evaluate the impact of local hotspots and heat removal, and thus to reveal battery runaway and propagation on cell and pack levels. The studies may be further extended to include the effects of different active materials and electrolytes, as reactions, reactivity and mechanical stability may change. Different experimental behaviours were reported here, especially for the highly reactive Li metal and solid state electrolytes. Also of special importance is the interaction of the cathode with water impurities and of metal dissolution on the reaction network and the thermal safety behaviour.

Abbreviations

a	Activity (—)
α	Dissociation degree (—)
$C_{p,bat}$	Battery heat capacity (J K^{-1})
d_{SEI}	SEI thickness (m)
E_A	Activation energy (J mol^{-1})

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$\Delta_r G$	Gibbs free energy of reaction (J mol^{-1})
$\Delta_r H_j$	Enthalpy of reaction (J mol^{-1})
$k_{f/b}$	Frequency factor forward/backward reaction (mol s^{-1}, (mol m) s^{-1})
K	Equilibrium constant (—)
ν	Stoichiometric coefficient (—)
n	Molar amount (mol)
r	Reaction rate (mol s^{-1})
R	Molar gas constant (J (mol^{-1} K^{-1}))
$\Delta_r S$	Entropy of reaction (J (mol^{-1} K^{-1}))
t	Time (h)
t_{SH}	Self-heating time (h)
t_{TR}	Thermal runaway time (h)
T	Temperature (K, °C)
T_{SH}	Self-heating temperature (K, °C)
T_{TR}	Thermal runaway temperature (K, °C)

Data availability

Source data are provided with this paper in the KITopen repository under **https://doi.org/10.35097/1804**.

Author contributions

Florian Baakes: conceptualisation, methodology, software, writing –original draft, validation, investigation, visualisation. Daniel Witt: conceptualisation, writing – review & editing. Ulrike Krewer: conceptualisation, writing – review & editing, supervision, funding acquisition.

Conflicts of interest

There are no conflicts to declare.

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